AMRITA VISHWA VIDYAPEETHAM
श्रद्धावान् लभते ज्ञानम्
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SellSmart: A Risk-Aware Sell-Decision Advisor for Small Farmers

Team ID: *GEN26-A0H-I154*

Team Name: *NEXUS0*

Domain: *Food Security, Agriculture & Rural Transformation*

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Presentation Outline

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Problem Statement

- **Uninformed Selling Decisions:** Farmers often sell their harvested crops at the nearest mandi or familiar market without comparing other available options.
- **Price Does Not Equal Profit:** A higher market price may still result in lower earnings after transportation costs, market charges and possible crop spoilage.
- **Unpredictable Market Prices:** Daily price fluctuations make it difficult to decide whether to sell immediately or wait for a potentially better price.
- **Limited Resources:** Small farmers may lack storage facilities, affordable transportation and the flexibility to delay selling due to urgent financial needs.
- **Lack of Personalized Guidance:** Existing price information alone may not account for a farmer's crop quantity, selling urgency, travel distance and risk tolerance.
- **Our Problem Statement:** How can we help small farmers identify the most suitable market and selling time by comparing expected net earnings, costs and risks according to their individual needs?

Problem Understanding

- **Target Users:** Small and marginal farmers growing perishable crops like tomato and onion, who often need to sell soon after harvest.
- **Price vs. Profit:** A mandi offering a higher price may not give higher earnings after transport costs, market charges and spoilage are deducted.
- **Uncertain Selling Time:** Farmers struggle to decide whether to sell today or wait, as future prices may rise or fall while their crops lose quality.
- **Personal Constraints:** Urgent cash needs, limited storage and travel distance make some selling options impractical.
- **Core Problem:** Farmers lack a simple, personalized way to compare where and when to sell based on actual net earnings, risk and their own limitations.

Existing Gaps and Challenges

- **Price-Focused Decisions:** Available market information often highlights crop prices, but farmers also need to understand the actual profit after selling-related expenses.
- **Scattered Information:** Market prices, transportation costs and other selling expenses may need to be checked separately, making comparisons difficult.
- **No Personalized Comparison:** General market prices may not consider each farmer's crop quantity, location, urgency and available resources.
- **Uncertain Future Prices:** Farmers cannot know with certainty whether prices will increase or decrease, making decisions about when to sell challenging.
- **Hidden Losses and Risks:** Transportation expenses, delays and deterioration of perishable crops can reduce the money a farmer finally receives.
- **Decision-Making Gap:** Farmers need a simple way to compare feasible markets and selling options based on expected net earnings, risks and personal constraints.

Our Proposed Solution

What is SellSmart?

SellSmart is a proposed software-based, risk-aware sell-decision advisor that helps small farmers compare where and when to sell their crops using estimated net earnings, practical constraints and price uncertainty.

- **Compare markets:** Identify candidate mandis using available prices, distance and farmer-defined travel limits.
- **Calculate net earnings:** Estimate the value of saleable produce after transport, market charges, handling expenses and applicable spoilage assumptions.
- **Evaluate selling time:** Compare selling now with waiting only when storage, crop condition and the required data make that option feasible.
- **Account for risk:** Present uncertainty and possible downside instead of treating a predicted price as certain.
- **Explain recommendations:** Show why an option ranks higher, including its estimated return, major costs and important limitations.

Technical Approach

From Farmer Inputs to Risk-Aware Recommendations

1. INPUT
Crop, quantity, location and urgency

→

2. ANALYSE
Prices, transport costs and spoilage risk

→

3. RANK
Compare net earnings and farmer constraints

→

4. RECOMMEND
Suitable markets with estimated profit and risk

Technology Stack

HTML/CSS/JavaScript — Interface • **Python + Flask** — Backend
Pandas — Data analysis • **SQLite** — Prototype database

Innovation: Risk-Aware Decision Engine

Expected Net Earnings = Expected Sale Value Estimated Selling Costs

Rank feasible options using spoilage risk, price uncertainty and farmer needs.

Innovation and Uniqueness

From Price Comparison to Smarter Selling Decisions

1. Profit-First

Beyond Market Price

Compares expected earnings after transport, market charges and spoilage costs.

2. Risk-Aware

Beyond Today's Price

Considers price uncertainty, crop urgency and the risk of waiting.

3. Farmer-Centric

Beyond One-Size-Fits-All

Considers quantity, travel distance, storage and individual constraints.

What Makes It Unique?

Market Price + Selling Costs + Risk + Farmer Needs

$\Downarrow$

A Personalized Selling Recommendation

SellSmart recommends what may work best for the farmer, not simply which mandi offers the highest price.

Feasibility and Scalability

Build Small • Validate • Scale Smart

Technical Feasibility

- Python, Flask and SQLite.
- Available mandi-price data.
- Rule-based profit and risk analysis.
- Simple web-based prototype.

Cost Feasibility

- Free, open-source tools.
- Student-led development.
- Start with essential features.

Scalability Roadmap

- **Start:** Test with sample data.
- **Validate:** Use real market prices.
- **Expand:** Add crops and mandis.
- **Scale:** Reach farmers and FPOs.

Future Enhancements

- Multilingual mobile access.
- Updated market-price feeds.
- Historical-data-based risk estimates.

Cost of Product

Estimated Prototype Development Cost

SellSmart: A Risk-Aware Sell-Decision Advisor for Farmers

Development Cost

- UI Design and Frontend: Rs.2,000
- Backend and Decision Engine: Rs.3,000
- Data Processing and Testing: Rs.2,000
- Hosting and Miscellaneous: Rs.1,000

Estimated Total

Rs.8,000

Cost and Sustainability

- **Technology:** Python, Flask, HTML, CSS and SQLite.
- **Data:** Use available mandi-price datasets or verified sources.
- **Operating Cost:** Depends on hosting, data access and usage.
- **Target Users:** Small and marginal farmers, farmer groups and FPOs.
- **Future Model:** Basic access for farmers, with optional services for FPOs and organisations.

Use Cases

- **Use case 1 – Compare mandis:** A tomato farmer with 10 quintals compares a local mandi at INR 2,400/quintal with a mandi 40 km away at INR 2,600. In the illustrative example, SellSmart estimates INR 25,080 net at the second mandi versus INR 23,370 locally, after assumed costs.
- **Use case 2 – Urgent sale, no storage:** An onion farmer needs cash today and has no storage. SellSmart removes waiting options and compares only reachable mandis, reducing the chance of recommending an impractical delay.
- **Use case 3 – FPO adviser (planned pilot):** An FPO adviser compares options for farmers with different travel limits and tests how a rise in transport cost or a price fall changes recommendations.

Impacts and Benefits

Economic Benefits

- Compare market prices and selling costs.
- Identify options with better expected net earnings.
- Reduce the risk of unprofitable selling decisions.

Farmer Empowerment

- Consider urgency, distance and storage.
- Make informed, personalized decisions.
- Support small and marginal farmers.

Practical Impact

- Understand transport and spoilage risks.
- Compare suitable markets easily.
- Extend support to farmer groups and FPOs.

Expected Outcome

Better-informed selling decisions through cost, profit and risk comparison.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.3	Zero Hunger	Supports small-scale farmers by helping them make better-informed selling decisions and work toward improved income.
SDG 12	Target 12.3	Responsible Consumption and Production	Better selling-time decisions may help avoid some post-harvest loss; this impact must be measured in a pilot.

Individual Member Contribution

Member	Contribution or responsibility
Hashika N	Research public mandi-price sources and clean the selected crop data.
Muskaan Chhabra	Develop the net-earning formula, transport-cost table and mandi filter.
Madana Ankitha	Work on forecasting, uncertainty, risk scoring and back-testing.
Uthara U	Build the Streamlit interface, What-If controls and presentation / testing materials.

References

- Mahajan, P., Muley, A. A., & Fegade, M. R. (2026). *Forecasting agricultural commodity prices using machine learning models*. Discover Informatics, 1, Article 15. [DOI]
- Manogna, R. L., Dharmaji, V., & Sarang, S. (2025). *Enhancing agricultural commodity price forecasting with deep learning*. Scientific Reports, 15, 20903. [DOI]
- Aker, J. C. (2011). *Dial "A" for agriculture: A review of information and communication technologies for agricultural extension in developing countries*. Agricultural Economics, 42(6), 631–647. [DOI]
- Aker, J. C. (2010). *Information from markets near and far: Mobile phones and agricultural markets in Niger*. American Economic Journal: Applied Economics, 2(3), 46–59. [DOI]
- Fafchamps, M., & Minten, B. (2012). *Impact of SMS-Based Agricultural Information on Indian Farmers*. The World Bank Economic Review, 26(3), 383–414. [DOI]

Data sources: AGMARKNET (daily mandi prices and arrivals); CEDA Agri-Market Data Portal (historical agricultural market data). *The papers motivate market-information and price-forecasting methods; SellSmart's own performance must be established through its back-test.*

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